CSA4704-DEEP LEARNING

NAME:HARITHI S **REG NO:192421151**

Experiment 10: L2, Dropout and Early Stopping Comparison
Problem Statement: A medical-records team trained a large MLP that scores 99% on training data but performs poorly on new patients. A small tabular dataset (around 500 records) for predicting diabetes onset from clinical features. The objective is to compare L2 regularisation, dropout and early stopping as remedies for overfitting. Use TensorFlow, Keras, NumPy, Matplotlib.
Tasks:
1. Train a deliberately large baseline MLP and plot its overfitting.
2. Retrain with L2 regularisation at three λ values.
3. Retrain with dropout at rates 0.2, 0.3 and 0.5.
4. Retrain with early stopping using validation loss.
5. Tabulate test accuracy and train–test gap for every model.

FORMULA:

1. L2 Regularisation

$L_total = L_data + \lambda \Sigma w_i^2$

2. Dropout

$h_i' = h_i \times m_i$

where m_i is a random binary mask applied to hidden units during training, with inverted dropout scaling.

3. Binary Cross-Entropy Loss

$L = -(1/N) \Sigma [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$

4. Early Stopping

Stop training when the validation loss does not improve for a specified number of epochs (patience).

5. Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN) × 100%

6. Training–Test Accuracy Gap

Gap = Training Accuracy − Test Accuracy

7. Overfitting

Overfitting occurs when the model performs well on training data but poorly on unseen test data.

8. L2 Regularisation Values

$\lambda \in \{0.001, 0.01, 0.1\}$

9. Dropout Rates

Dropout Rate $\in \{0.2, 0.3, 0.5\}$

CODE:

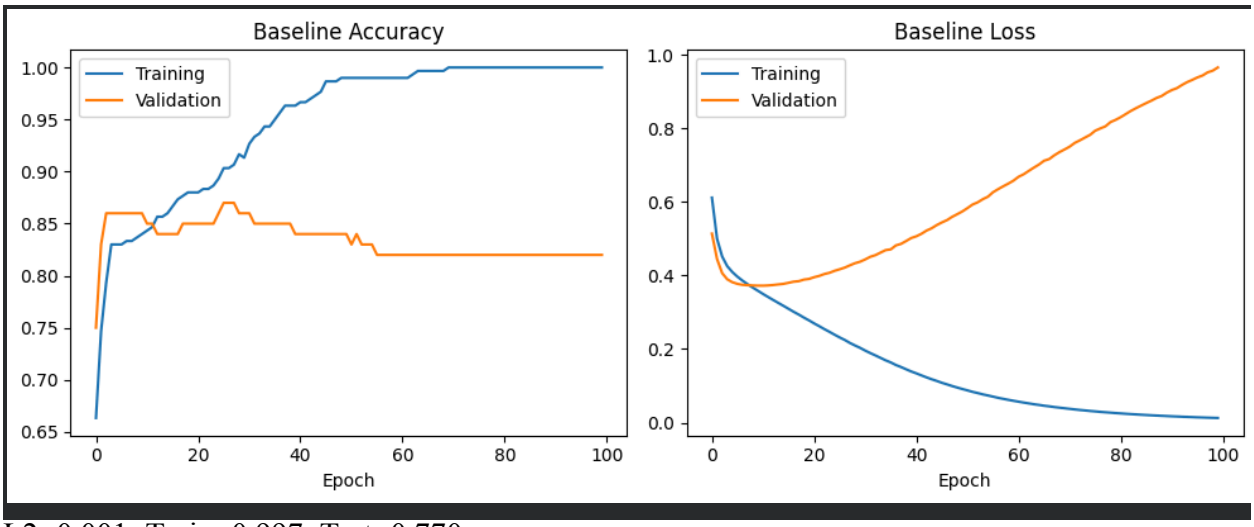
```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras import Sequential, regularizers
from tensorflow.keras.layers import Input, Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping

np.random.seed(42)
tf.random.set_seed(42)

…plt.bar(names, test_scores)
plt.ylabel("Test Accuracy")
plt.title("Regularisation Method Comparison")
plt.xticks(rotation=35, ha="right")
plt.tight_layout()
plt.show()
```

OUTPUT:

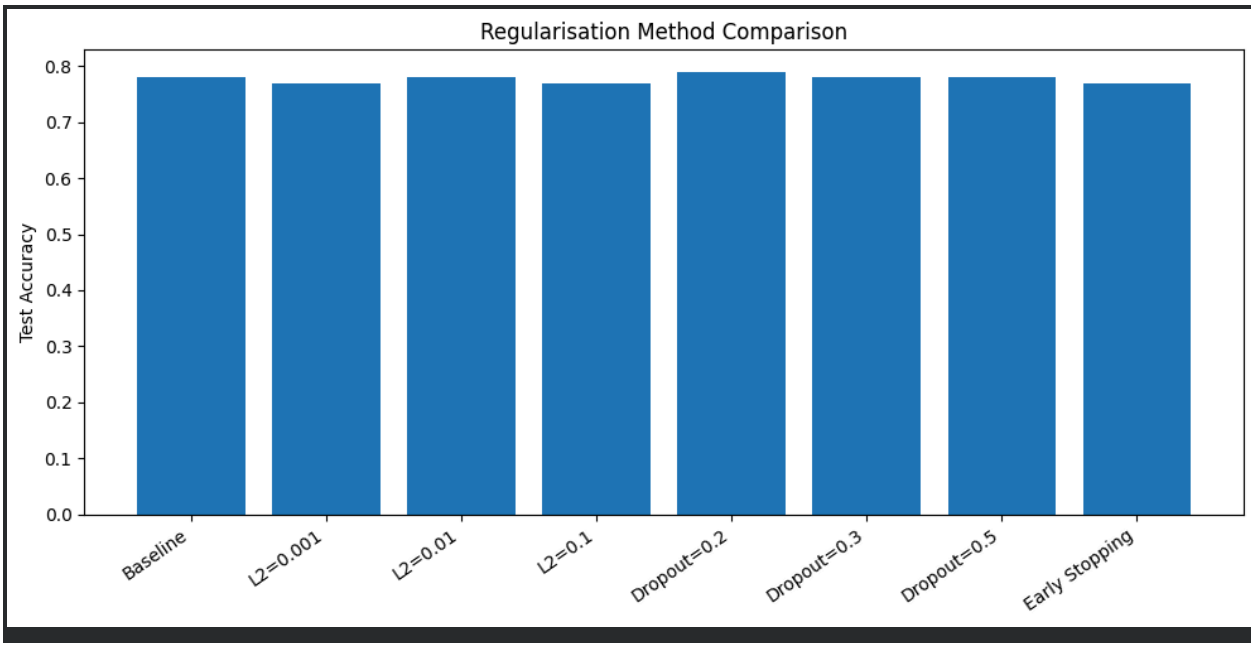
```
Baseline: Train=1.000, Test=0.780
```



L2=0.001: Train=0.997, Test=0.770
L2=0.01: Train=0.910, Test=0.780
L2=0.1: Train=0.823, Test=0.770
Dropout=0.2: Train=0.990, Test=0.790
Dropout=0.3: Train=0.970, Test=0.780
Dropout=0.5: Train=0.920, Test=0.780
Early Stopping: Train=0.860, Test=0.770

FINAL COMPARISON

Model	Train Acc	Test Acc	Gap	Epochs
Baseline	1.000	0.780	0.220	100
L2=0.001	0.997	0.770	0.227	100
L2=0.01	0.910	0.780	0.130	100
L2=0.1	0.823	0.770	0.053	100
Dropout=0.2	0.990	0.790	0.200	100
Dropout=0.3	0.970	0.780	0.190	100
Dropout=0.5	0.920	0.780	0.140	100
Early Stopping	0.860	0.770	0.090	20


Regularisation Method Comparison
Test Accuracy
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Baseline
L2=0.001
L2=0.01
L2=0.1
Dropout=0.2
Dropout=0.3
Dropout=0.5
Early Stopping